AI Assignment 2 | UAV Strategy Engine

Bahria University, Lahore Campus | BSCS 5 | Artificial Intelligence

Team members (3): _____ / _____ / _____

Objective. Design a Genetic Algorithm (GA) that assigns a three-UAV fleet to high-priority disaster cells while reducing sensor/flight resource consumption and avoiding hazardous or restricted regions. The simulation uses a 30 x 30 two-dimensional synthetic grid, with four scans per drone and a home base at (1, 1).

1. Hybrid chromosome encoding

A chromosome contains three NumPy arrays, indexed by UAV d and scan stop k. The **waypoint gene** is (x[d,k], y[d,k]) with real coordinates in [0,29]; **speed[d,k]** is continuous in [0.8,2.0] simulated distance/time units; and **sensor[d,k]** is integer 0 (off), 1 (low), or 2 (high). Thus one candidate represents all deployment positions, trajectories and scan power settings. Each route begins and ends at base. Example: UAV1 = waypoints [(5,8), (12,7), (20,15), (25,22)], speeds [1.2,1.5,1.0,1.8], sensors [2,1,2,1].

Design rationale. Real-valued coordinates permit smooth geometric mutation, speed controls the time/energy trade-off, and discrete sensor settings represent a practical scheduling decision. Segments can be exchanged between parents without mixing different UAVs. A repair step confines waypoint coordinates to legal grid locations.

2. Multi-objective fitness function

Maximize: F = 0.32 C + 0.20 D - 0.17 E - 0.13 T - 0.18 R - 2.0 V, where C is the priority-weighted fraction of cells within at least one active scan footprint, D is priority-weighted detection probability proxy, E = min(energy/450,1), T = min(aggregate_time/240,1), R = min(hazard_integral/130,1), and V counts residual constraint violations. The five objective weights sum to 1; the penalty is purposely larger than any objective gain.

Coverage weights high-priority disaster cells more heavily than low-priority cells. D combines repeated scan probabilities using 1 - product(1 - individual_scan_strength). Energy is a proxy depending on route length, speed squared, and sensor power; time uses distance/speed plus sensor scan duration; risk integrates hazard intensity along route samples. These are simulation assumptions, not physical UAV specifications.

3. Evolution mechanics and safety constraints

Initialize random populations, evaluate fitness, select parents by tournament (default size 3), exchange one entire UAV plan, geometrically blend a second waypoint, and mutate waypoint coordinates (Gaussian shifts or random relocation), speed, and sensor settings. Retain elite candidates. Stop after 20 generations. Hard constraints include rectangular no-fly zones, a maximum distance of 105 units per UAV, and fleet flight-time cap of 225 units. Waypoint repairs use nearest free cell; a heuristic axis-aligned detour is attempted where a straight segment intersects a no-fly zone. Unresolved intersections, overlength routes, or excessive duration receive significant penalties.

Results | Convergence and Route Evolution

Experiment: seed 12, population 18, 20 generations, mutation rate 0.14, tournament size 3. The random-search baseline used the same total candidate-evaluation budget (18 x 21 = 378).

GA: fitness 0.0477, weighted coverage 57.5%, detection 42.8%, energy 255.9, time 156.0, risk 29.5, constraint violations 0.

Random search: fitness -0.1268, weighted coverage 30.1%, detection 20.6%, energy 274.2, time 168.2, risk 50.2, constraint violations 0.

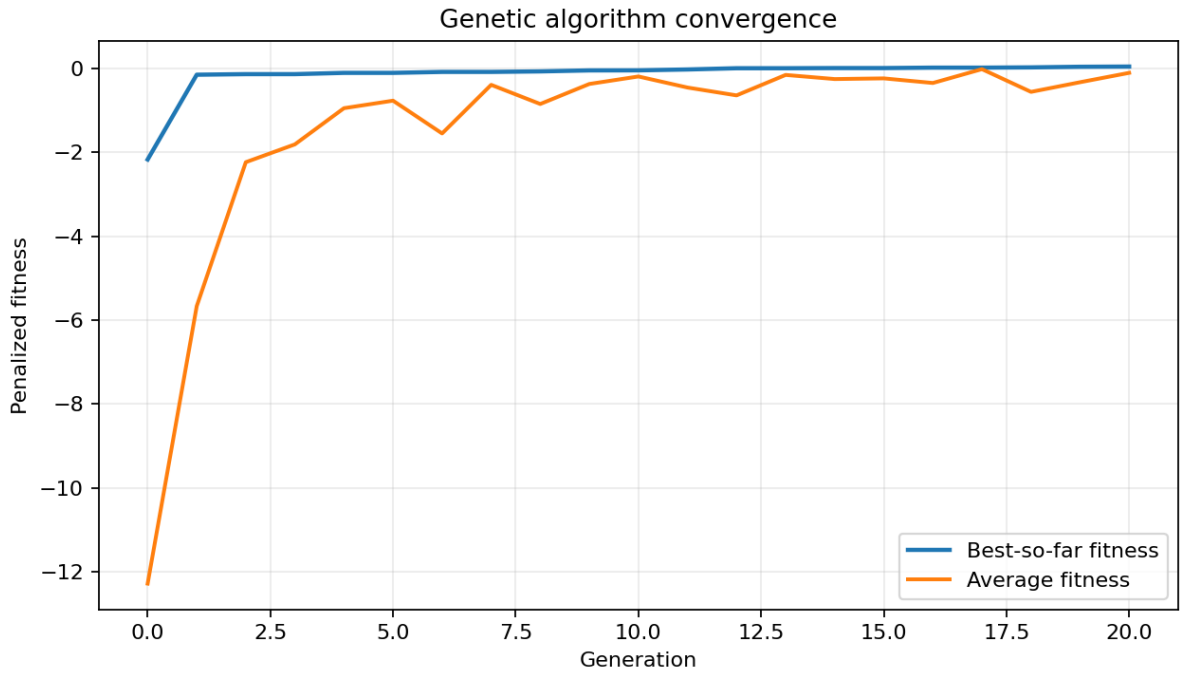


Figure 1. Best-so-far fitness versus average population fitness across generations.

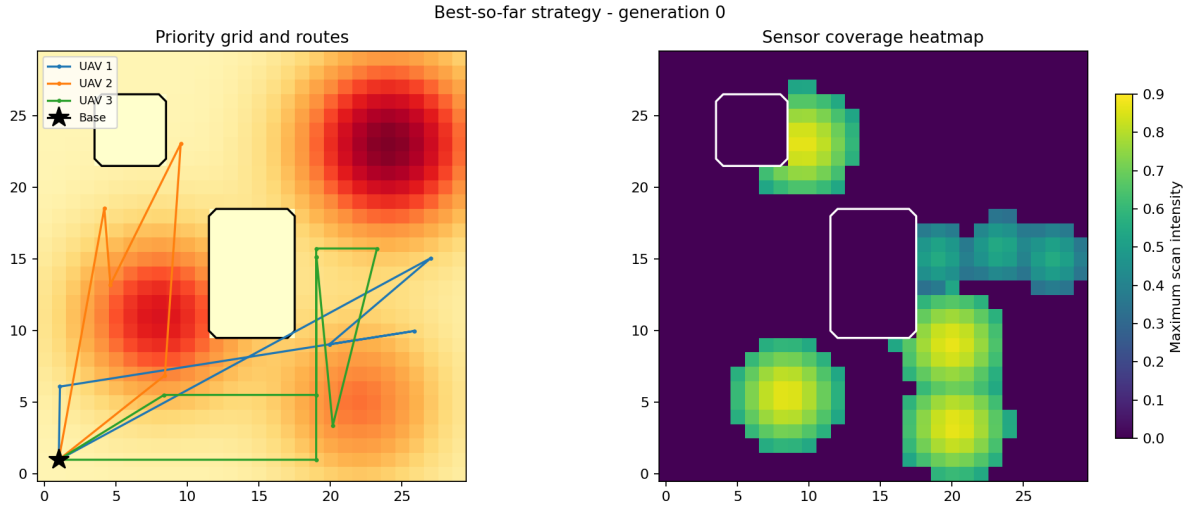


Figure 2. Early-generation flight paths (left) and estimated scan coverage (right).

Results | Evolved Flight Paths

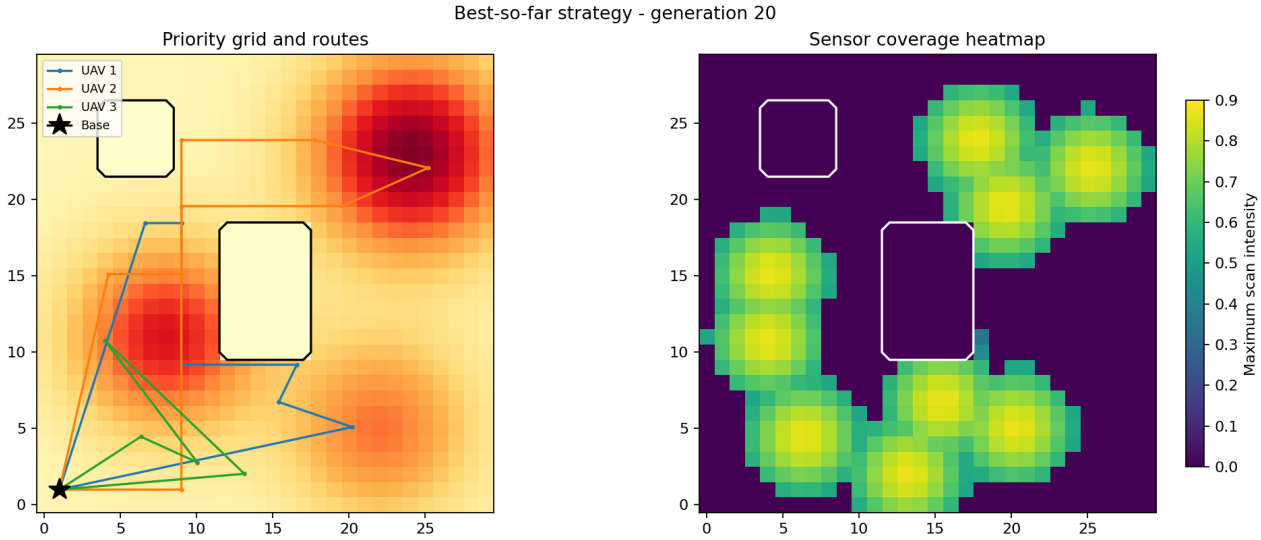


Figure 3. Final best-so-far flight paths and coverage heatmap (generation 20).

4. Comparison against random search

The GA final candidate is stronger than the random-search baseline under the same scoring function and 378 evaluated candidates. The GA uses tournament selection, inheritance, and mutation to improve known solutions; random search samples unrelated strategies. For this one seed, GA fitness exceeds random fitness, while weighted coverage and target-detection proxies also improve. Neither approach proves a global optimum.

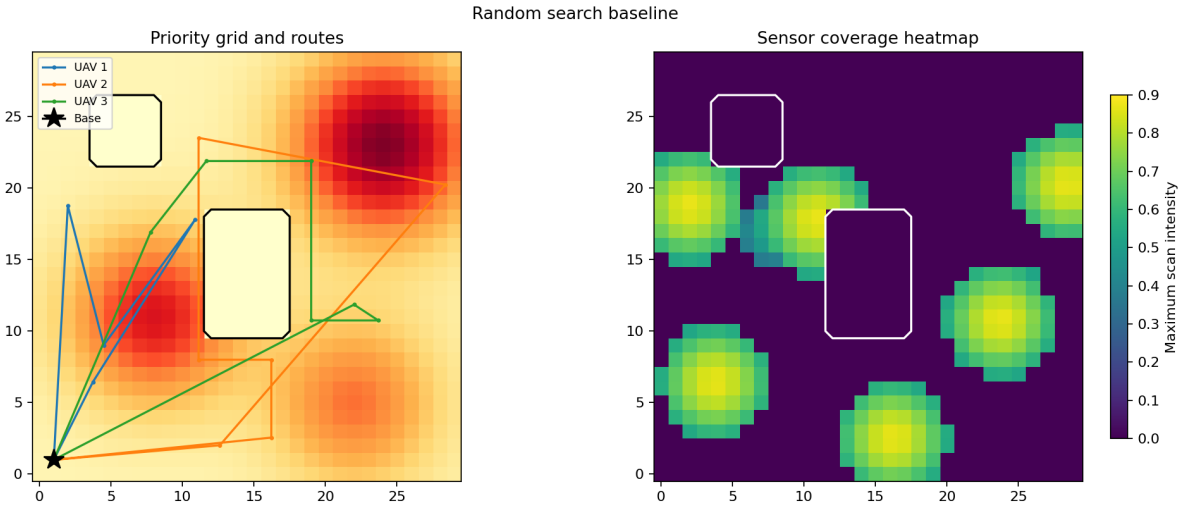


Figure 4. Best random-search baseline path and heatmap under matched evaluation count.

Hyperparameter Sensitivity and Conclusions

5. Hyperparameter sensitivity analysis

The following are one-seed illustrative experiments with 10 generations per run, changing one parameter while retaining the others near the default. Because varying the population also varies the number of candidate evaluations, results indicate possible trends but are not controlled evidence of a population-size effect. Multiple independent seeds are required for statistical confidence.

Parameter	Value	Best fitness	Weighted coverage	Violations
population	12	-0.0391	0.326	0.0
population	24	-0.0181	0.382	0.0
mutation	0.05	-0.1337	0.313	0.0
mutation	0.28	-0.0389	0.350	0.0
tournament	2	-0.0577	0.409	0.0
tournament	5	-0.0588	0.550	0.0

Interpretation. Larger populations can expose more strategies at higher computational cost. Too little mutation may trap the search in local patterns; too much can disrupt inherited good routes. Stronger tournament selection emphasizes high-fitness parents but can reduce genetic diversity. Specific numeric results depend on seed and the chosen hazard map, so avoid over-generalizing differences between settings.

6. Why the evolved solution is distinct

The GA is spatially and operationally aware. Drone-level segment swaps preserve coherent assignments, geometric coordinate blends exploit nearby route alternatives, and sensor/speed genes encode trade-offs that random search does not exploit iteratively. The selected final paths may differ across seeds. The method seeks one weighted compromise, rather than a complete Pareto frontier.

7. Limitations and recommended extensions

Model limitations include approximate flight physics, static hazard maps, simplified detection probabilities, waypoint rather than continuous scan planning, limited detour candidates, and one-seed sensitivity results. An upgraded engine should optimize non-dominated solutions (e.g., NSGA-II), test multiple random seeds, enforce per-UAV energy budgets, model sensor field-of-view and time-varying weather, and use A* for more complete obstacle-aware path repair.

8. Execution and reproducibility

Install requirements using `pip install -r requirements.txt`. Run `python strategy_engine.py --output outputs --seed 12 --population 18 --generations 20 --mutation 0.14`. The script exports fitness_history.png and fitness_history.csv, route maps, coverage maps, random baseline results, and a sensitivity CSV. Refer to README.md for file layout and methodology. Repository publication to GitHub must be performed by the team; this deliverable is GitHub-ready but not uploaded to GitHub.

Academic note. All scores are generated by an educational model using synthetic map values. They are not measured disaster response performance or evidence of safe physical drone deployment.